Mathematical Evidence Registry v0.1.1

Kelvin spectra, conditional gap thermodynamics,
and conservative versus effective two-sector dynamics

Independent mathematical audit of historical SST manuscripts

Prepared 21 October 2026

DEVELOPMENT — NOT PHYSICALLY CERTIFIED

Abstract

This release adds 22 source-traceable findings to SST Mathematical Evidence Registry v0.1.0 without changing its 379 unreviewed raw formula extractions, 41 previously curated claims, or source fingerprints. It independently examines two Kelvin-gap manuscripts and a proposed hydrodynamic dual-vacuum model. Mathematical validity, independently measured vortex dynamics, quantum interpretation, and empirical evidence are strictly separated. A Python standard-library validation suite exercises 55 regression tests (23 retained and 32 new), including dispersive scaling, Bose oscillator thermodynamics, analytic rate models and a positive-energy conservative coupled oscillator. All physical identifications remain unresolved.

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1 Source provenance and epistemic constraints

The primary manuscripts are (i) *Kelvin Mode Suppression in Atomic Orbitals: A Vortex-Filament Gap*, 10 pages; (ii) *Suppression of Kelvin-Mode Thermodynamics in Atomic Orbitals*, 8 pages (historical preprint dated 14 December 2025); and (iii) *Hydrodynamic Dual-Vacuum Unification*, 5 pages (1 December 2025). The first two share essentially the same proposed mechanism and are not independent evidence. Full source byte hashes appear in `results/source_manifest.json`. All cited page numbers are physical PDF pages, not automatically assigned equation IDs.

Provenance policy. An equation that is mathematically correct given explicit assumptions may be retained, without validating the assumption or its proposed interpretation. A result calculated from a supplied circulation, a supplied frequency, or a chosen gap is not an independent prediction of that input. Historical summaries and prior AI assessments are discovery indexes, never primary proofs.

2 Kelvin-wave dispersion and circulation incompatibility

2.1 Asymptotic dispersion: mathematically useful

For small bending disturbances of a locally straight slender vortex filament, a commonly used long-wave relation is [1, 2]

$$\omega_{\mathrm{loc}}(k) = \frac{\Gamma k^2}{4\pi}\left[\ln\frac{1}{|k|\xi} + C_0\right], \quad |k|\xi \ll 1, \tag{1}$$

where circulation $[\Gamma] = \mathrm{m}^2\,\mathrm{s}^{-1}$, wavenumber $[k] = \mathrm{m}^{-1}$, core length $[\xi] = \mathrm{m}$ and the dimensionless constant C_0 depends on the core prescription. Hence $[\omega] = \mathrm{s}^{-1}$. The periodic surrogate $k_m = 2\pi m/L$ gives a diagnostic scale, not the actual eigenvalues of a curved or knotted loop. In particular, the removal of global translation/rotation and circulation-preserving constraints is necessary before interpreting low-m branches as intrinsic modes.

The first Kelvin manuscript uses $\Gamma_{\mathrm{paper}} = h/m_e$, whereas the current SST canon defines $\Gamma_{\mathrm{canon}} = 2\pi r_c v_{\mathrm{swirl}}$. These are incompatible numerical prescriptions unless a distinct circulation sector is explicitly established:

$$\Gamma_{\mathrm{paper}} = 7.27389509 \times 10^{-4}\ \mathrm{m}^2\mathrm{s}^{-1}, \tag{2}$$

$$\Gamma_{\mathrm{canon}} = 9.68361920 \times 10^{-9}\ \mathrm{m}^2\mathrm{s}^{-1}, \tag{3}$$

$$\frac{\Gamma_{\mathrm{paper}}}{\Gamma_{\mathrm{canon}}} = 75\,115.4598. \tag{4}$$

Using the same geometry in Eq. (1), all frequencies scale directly with the chosen Γ. This is a change of input, not a new physical effect. The printed frequency $\omega_1 = 2.8 \times 10^{17}\,\mathrm{s}^{-1}$ corresponds to $\hbar\omega_1 = 184.299\,\mathrm{eV}$, not 29.5 eV; both figures appear in the original manuscript's appendix (p. 8).

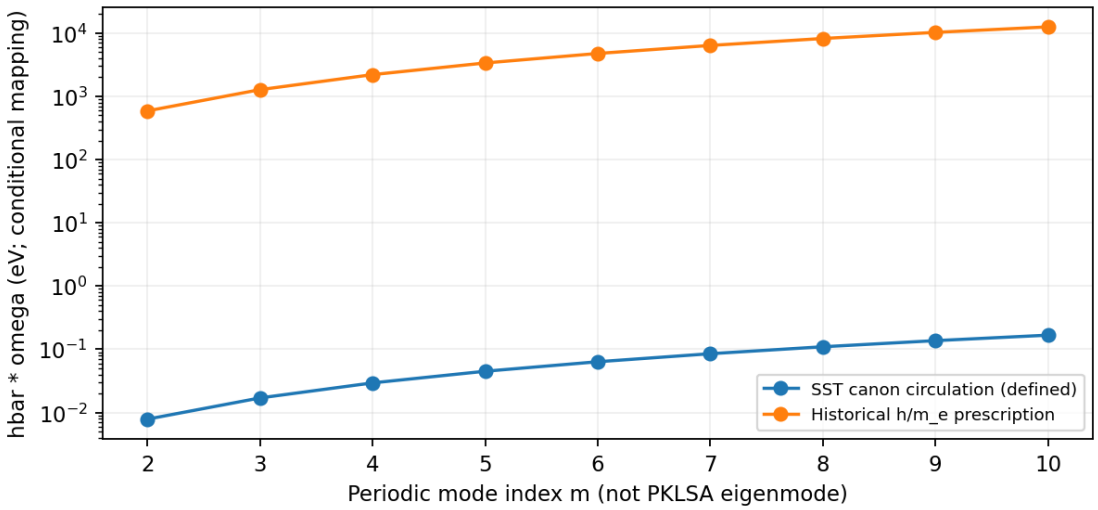


Figure 1: Illustrative straight-filament/periodic-wave surrogate at fixed $L = 2\pi a_0$, $\xi = r_c$, $C_0 = 1$; $m \geq 2$. **Not** the spectral output of Euler/PKLSA, and $\hbar\omega$ is an imported energy mapping.

2.2 Finite-length spacing is not an independently derived topological gap

A bounded periodic arclength may produce a discrete sequence of k_m. It does not by itself prove a universal positive *topological* excitation gap; symmetry-related zero modes and curvature effects need separate treatment. The correct native test computes the linearized operator around an actual stationary/relative-periodic vortex, removes symmetry-neutral directions, and checks its eigenvalues under spatial and time refinement. A positive minimal frequency of a classical linearized system also does not independently establish a quantum excitation energy $\hbar\omega$.

2.3 Dimensional failures in the proposed gap estimate

The first manuscript's $\Delta_K \sim \Gamma^2/(4\pi\xi L)$ has dimensions

$$\left[\Gamma^2/(\xi L)\right] = \mathrm{m}^2\mathrm{s}^{-2},$$

a *specific* energy scale, not joules. The putative tension $\rho_f\Gamma^2/(4\pi\xi^2)$ has units $\mathrm{N\,m^{-2}}$ (pressure), not N (energy per unit filament length). The actual line energy of a slender vortex typically involves $\rho_f\Gamma^2$ multiplied by a dimensionless logarithmic factor. A mode-specific effective inertia and finite-core dynamical model are required before any absolute gap estimate.

3 What the Kelvin-gap thermal mathematics establishes

3.1 Single-mode partition function

Assume an *independent quantized bosonic harmonic mode* with excitation energy $\Delta > 0$ and inverse temperature $\beta = (k_B T)^{-1}$, with zero-point energy subtracted. Then [3]

$$Z_{\mathrm{exc}} = \sum_{n=0}^{\infty} e^{-n\beta\Delta} = \frac{1}{1-e^{-\beta\Delta}}, \tag{5}$$

$$U_{\mathrm{exc}} = -\frac{\partial}{\partial\beta}\ln Z_{\mathrm{exc}} = \frac{\Delta}{e^{\beta\Delta}-1}, \tag{6}$$

$$C = \frac{\partial U}{\partial T} = k_B \frac{x^2 e^{-x}}{(1-e^{-x})^2}, \quad x = \frac{\Delta}{k_B T}. \tag{7}$$

For $x \gg 1$, $U_{\mathrm{exc}} \sim \Delta e^{-x}$ and $C \sim k_B x^2 e^{-x}$. The result **does not derive** Δ from vortex geometry, and a classical inviscid closed Euler fluid has no specified thermal bath, statistical state, or quantization rule.

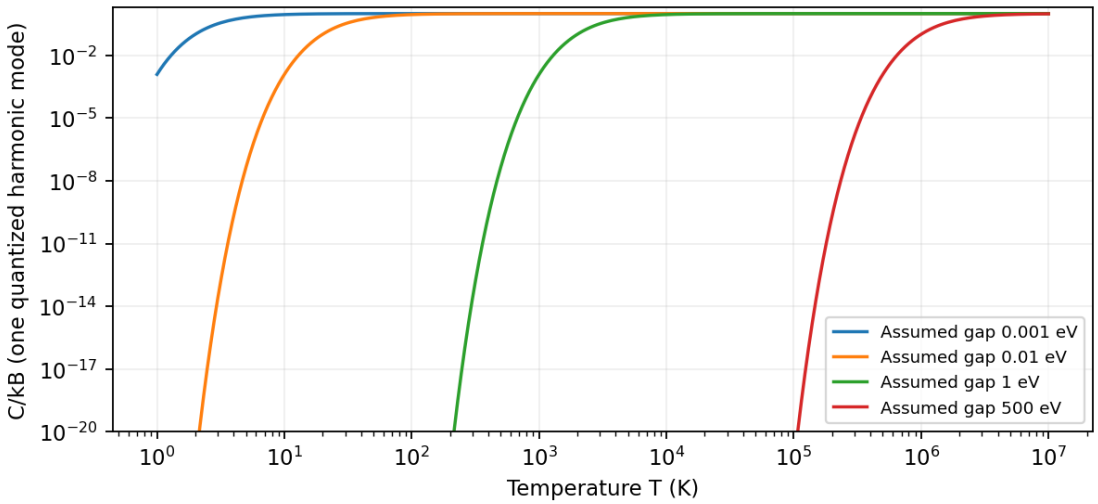


Figure 2: Heat capacity predicted after *assuming* independent quantized oscillator gaps. This does not establish either the existence or value of a Kelvin gap.

If there are finitely many independent oscillators $j = 1, \ldots, N$ with energies $\Delta_j \geq \Delta_* > 0$, then monotonicity of $f(\Delta) = \Delta/(e^{\beta\Delta}-1)$ yields the exact conditional bound

$$0 \leq U_{\mathrm{total}} \leq N \frac{\Delta_*}{e^{\Delta_*/(k_B T)}-1}. \tag{8}$$

The manuscript's $N\Delta_* e^{-\Delta_*/k_B T}$ is a low-temperature asymptotic estimate, not a strict upper bound at finite temperature.

The condition $\Delta_*/(k_B T_{\mathrm{eff}}) \geq 60$ at $T_{\mathrm{eff}} = 10^5\,\mathrm{K}$ gives $\Delta_* \geq 517.040\,\mathrm{eV}$. The dimensionless threshold 60, effective temperature and claimed spectroscopy bound $a_n \leq 10^{-62}\,\mathrm{J\,K^{-2}}$ are external modeling premises lacking an independently reconstructed measurement contract in these source PDFs. A T^2 correction is also not a universal prediction for an arbitrary finite set of Kelvin oscillators.

3.2 The Schrodinger variational identity is conditional

The papers insert the energy functional

$$\mathcal{F}[\psi] = \int d^3x \left[\frac{\hbar^2}{2m_e}|\nabla\psi|^2 + V(\mathbf{x})|\psi|^2\right], \quad \int |\psi|^2 d^3x = 1.$$

Its stationary variation yields $-\hbar^2\nabla^2\psi/(2m_e) + V\psi = E\psi$, a correct conditional result. Since the quantum gradient coefficient and potential were inserted, this does not derive quantum mechanics from the Euler flow. The status in the registry is VARIATIONAL_IDENTITY_ONLY.

4 Two-sector field dynamics: two distinct mathematical baselines

4.1 Repair of the dissipative rate model

The dual-vacuum manuscript (p. 4) adds $\kappa_{se} = 4Z_s/Z_{\text{bound}}$ to inverse lifetimes. Since both Z values have impedance units (Rayl), κ_{se} is dimensionless; adding it to rates in s^{-1} is invalid. Define a new transfer rate $g\,[\text{s}^{-1}]$ and an efficiency $0 \le \eta \le 1$. An explicitly energy-accounted *phenomenological* model is

$$\begin{aligned} \dot{E}_s &= -(d_s + g)E_s, \\ \dot{E}_e &= \eta g E_s - d_e E_e, \\ \dot{E}_{\text{loss}} &= d_s E_s + (1-\eta) g E_s + d_e E_e. \end{aligned} \tag{9}$$

Thus $d(E_s + E_e + E_{\text{loss}})/dt = 0$, and all three energy pools remain nonnegative for nonnegative input data. For $E_s(0) = E_0$, $E_e(0) = 0$, $a = d_s + g$ and $b = d_e$,

$$E_e(t) = \eta g E_0 \frac{e^{-at} - e^{-bt}}{b - a}, \qquad t_{\text{peak}} = \frac{\ln(b/a)}{b - a}. \tag{10}$$

For $a^{-1} = 0.1\,\text{ns}$ and $b^{-1} = 30\,\text{ns}$, this gives $t_{\text{peak}} = 0.572286\,\text{ns}$: **not** a 30-ns delayed peak. A separate memory, propagation or gated-detection mechanism would need explicit construction.

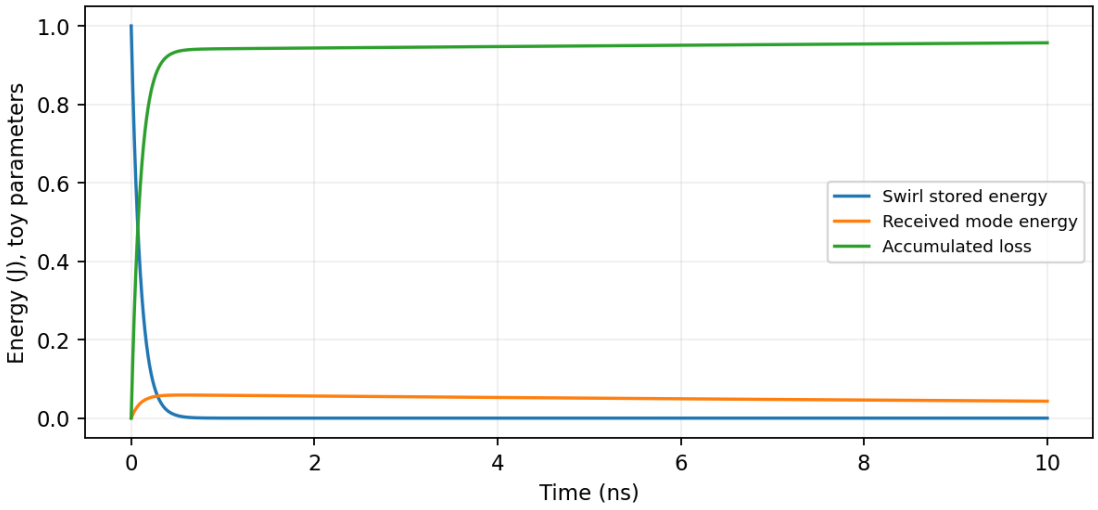


Figure 3: Energy-accounted *irreversible toy* rates with $g = 10^9\,\text{s}^{-1}$, $d_s = 9 \times 10^9\,\text{s}^{-1}$, $d_e = (30\,\text{ns})^{-1}$, $\eta = 0.6$, $E_0 = 1\,\text{J}$. Total accounted energy stays 1 J. No claim that these parameters describe nature.

4.2 Conservative coupled-mode action: independent construction

An equally useful, fully conservative *mathematical* model uses displacement-like variables q_s, q_e (metres), masses M_s, M_e (kg), stiffnesses k_s, k_e, g (N/m), and Lagrangian

$$L = \frac{M_s}{2}\dot{q}_s^2 + \frac{M_e}{2}\dot{q}_e^2 - \frac{k_s}{2}q_s^2 - \frac{k_e}{2}q_e^2 - \frac{g}{2}(q_s - q_e)^2, \tag{11}$$

$$H = \frac{p_s^2}{2M_s} + \frac{p_e^2}{2M_e} + \frac{k_s}{2}q_s^2 + \frac{k_e}{2}q_e^2 + \frac{g}{2}(q_s - q_e)^2, \qquad p_i = M_i\dot{q}_i. \tag{12}$$

$H \geq 0$ when all $M_i, k_i > 0$ and $g \geq 0$. The squared eigenfrequencies are the two eigenvalues of $M^{-1/2} K M^{-1/2}$ with

$$M = \begin{pmatrix} M_s & 0 \\ 0 & M_e \end{pmatrix}, \quad K = \begin{pmatrix} k_s + g & -g \\ -g & k_e + g \end{pmatrix}.$$

The included velocity-Verlet integrator checks conservative energy drift and second-order time convergence. **No** physical identity between q_e and the electromagnetic vector potential, or between q_s and PKLSA vorticity, is implied by this oscillator baseline. This baseline is separate from the coarse-grained rate equations.

4.3 Faraday coupling and constitutive closure

The source expression $-G\partial_t(\nabla \times \mathbf{v})\hat{\mathbf{n}}$ is not a well-defined vector without specifying a contraction/operator. One may *define* $\mathbf{B}_{\text{eff}} = \mathbf{B} + \chi\boldsymbol{\omega}$ with $[\chi] = \mathrm{T\,s}$. Then $\nabla \cdot \mathbf{B}_{\text{eff}} = 0$ follows from $\nabla \cdot (\nabla \times \mathbf{v}) = 0$. This is only a field redefinition. A genuine electromagnetic theory additionally needs independent dynamical/constitutive equations, gauge symmetry, energy exchange, and testable consequences. Likewise the characteristic ratio $c/v_{\text{swirl}} \approx 274$ is arithmetic; a tangential swirl speed is not automatically an independent signal speed.

5 Qualification gates for source-native PKLSA

Gate	Test	Current state
K0	Immutable provider, exact 3D seed and core model manifest	NOT CONNECTED
K1	True ring/trefoil equilibrium or relative periodic orbit	NOT RUN
K2	Linearized Kelvin/Floquet spectrum with symmetry modes projected out	NOT RUN
K3	Spatial $N = 1200, 2400$ and $4800, 9600$ plus independent time refinement	NOT RUN
K4	Extra gap versus finite-length mode spacing; source-native circulations	NOT RUN
K5	Optional quantization/thermal bath separately qualified	NOT RUN
D0	Conservative two-mode Hamiltonian numerical tests	MATH PASS
D1	Coarse-grained energy-rate positivity and dimensional tests	MATH PASS
D2	Source-native torsional mode / EM constitutive coupling	UNRESOLVED

6 Deliverables and validation meaning

Original `v0.1.0` outputs, raw extracted equations, and 11 source files are retained. This addendum contributes 22 claim records and independent numerical outputs (`kelvin_periodic_proxy.csv`, `conditional_thermal_grid.csv`, `two_sector_energy_rate_example.csv`, `conservative_oscillator_energy_trace.csv`). `run_all.cmd` rebuilds the entire index and addendum and executes 55 mathematical regressions. None is a native PKLSA or experimental certification.

References

References

[1] P. G. Saffman, *Vortex Dynamics*, Cambridge University Press (1992). DOI: 10.1017/CBO9780511624063.

[2] A. L. Fetter, "Rotating Trapped Bose–Einstein Condensates", *Rev. Mod. Phys.* **81**, 647–691 (2009). DOI: 10.1103/RevModPhys.81.647.

[3] R. K. Pathria and P. D. Beale, *Statistical Mechanics*, 3rd ed., Elsevier (2011). ISBN: 9780123821881.

[4] W. Thomson (Lord Kelvin), "Vibrations of a Columnar Vortex", *Philosophical Magazine* **10**, 155–168 (1880). Archive permalink.

[5] O. Iskandarani, "Suppression of Kelvin-Mode Thermodynamics in Atomic Orbitals: A Gapped Vortex-Filament Resolution in Swirl–String Theory", manuscript (14 December 2025), supplied PDF.

[6] O. Iskandarani, "Kelvin Mode Suppression in Atomic Orbitals: A Vortex-Filament Gap", supplied manuscript, 10 pp. (date not specified in this PDF).

[7] O. Iskandarani, "Hydrodynamic Dual-Vacuum Unification: Swirl Superradiance, Photon Torsion, and the Unruh Echo Mechanism", manuscript (1 December 2025), supplied PDF.